

Android Interview Prep Checklist

Senior Android Developer (11+ yrs) — Core Platform Topics

Architecture & System Design

1. Android Architecture and system design (overall app architecture)
2. MVVM with Clean Architecture
3. Common design patterns in Android — Repository, Factory, Observer, Singleton, Builder — with real examples from your projects
4. Mobile system design — "design an offline-first app" / "design a chat/sync system" style open-ended questions
5. Room database — migrations, relations, DAO patterns, Flow-backed queries
6. Navigation component — deep links, back stack handling, safe args
7. Jetpack Compose vs XML Views — recomposition, state hoisting, migration strategy

Lifecycle & Process Management

1. Activity / Fragment / Thread lifecycle
2. Process lifecycle & priority — process death/recreation, onSaveInstanceState / SavedStateHandle, cold/warm/hot start
3. ViewModel & SavedStateHandle — surviving config changes vs process death
4. WorkManager — deferrable/guaranteed background work, constraints, chaining

Core Components & IPC

1. Types of Services — Foreground, Background, Bound; System Service and its lifecycle
2. Broadcast Receivers — static vs dynamic, ordered broadcasts, local broadcasts
3. Intents — explicit vs implicit, intent filters, pending intents
4. AIDL — defining interfaces, marshalling, use cases
5. IPC communication — Binder, Messenger, ContentProvider, shared memory
6. ContentProviders — custom providers, URI matching, when to use vs other IPC
7. Permissions — runtime permissions, special permissions (overlay, all-files-access), scoped storage
8. Notifications — channels, foreground service notifications, importance levels
9. Foreground vs Background service restrictions (Android 8+, 12+ changes)

Memory & Performance

1. Memory management
2. How do you avoid memory leaks
3. ANR — causes, detection, StrictMode
4. App startup optimization — baseline profiles, lazy init, App Startup library
5. APK size & build optimization — R8/ProGuard, resource shrinking, multi-APK/App Bundles
6. Profiling tools — Android Profiler, LeakCanary, Perfetto/Systrace

Concurrency & Native

1. Multithreading and thread related questions
2. NDK + native libs integration

Security & Testing

1. Security — Keystore, encryption at rest, certificate pinning
2. Testing — unit tests (JUnit/Mockito), Espresso/Compose UI testing, testability-driven architecture

Note: Kotlin internals, Coroutines/Flows, and Dependency Injection (Hilt) to be revised separately.